

Hydro-PI-GNN: A Parametric Physics-Informed Graph Neural Network for Estuarine Hydrodynamics

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Dedication

To my amazing so and so...

and

To my beloved..., infinitely supportive

List of Publications

Journal Article

Publication 1: Authors of the paper, “Title of the paper”, Journal Name, Volume, Issue, Pages, Publication month and year, DOI

Conference

Publication 2: Authors of the paper, “Title of the paper”, Conference Name, Location, Date of Conference, Pages, DOI

Book Chapter

Publication 3: Authors of the Chapter, “Title of the Chapter”, in Book Name, Editors Name, Publishers, Pages, Publication year, DOI

Approval/Declaration by the Supervisor(s)

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Acknowledgement

[Acknowledge here those helped and guided you to pursue your thesis goal. Give thanks and show gratitude who worked actively to assist you.]

Abstract

Accurate and rapid prediction of estuarine hydrodynamics is critical for coastal flood risk management, maritime navigation, and real-time operational forecasting. Traditional physics-based numerical solvers, while highly accurate, are computationally prohibitive for large-scale probabilistic scenario assessments due to the strict Courant number stability constraints of high-resolution numerical integration, which force extremely small simulation time steps. To address this profound computational bottleneck, this thesis introduces **Hydro-PI-GNN**, a novel parametric Physics-Informed Graph Neural Network surrogate model engineered to dramatically accelerate hydrodynamic simulations on complex, unstructured grids.

The proposed model was developed, trained, and validated on the Gironde estuary—the largest macrotidal estuary in Western Europe. To provide high-fidelity ground truth data, a rigorous Delft3D Flexible Mesh (Delft3D FM) numerical model comprising 36,271 unstructured finite-volume cells was independently constructed and calibrated. To overcome the phenomenon of “spectral bias” inherent in standard deep learning architectures—which routinely causes networks to fail in capturing high-frequency tidal oscillations—Hydro-PI-GNN integrates a mathematically tailored dual-bandwidth Random Fourier Feature (RFF) positional embedding.

Operating directly on the spatial graph derived from the Delft3D FM mesh, the neural network learns to emulate the complete 2D Shallow Water Equations (SWE). Physical consistency is rigidly enforced via an autograd-based composite loss function that evaluates mass conservation and momentum residuals, incorporating crucial Green–Gauss reconstructed bathymetric bed-slope corrections. The surrogate was optimized using a sophisticated multi-stage curriculum learning

strategy and rigorously evaluated using an 80/20 temporal train/test split over a 265-hour tidal cycle.

Validation results on the entirely unseen, held-out test dataset demonstrate that Hydro-PI-GNN achieves exceptional predictive accuracy for the full hydrodynamic state (water surface elevation and depth-averaged velocity vectors) across the entire estuarine domain. Crucially, it dynamically conditions these predictions on continuous variations in ocean tidal amplitude and upstream riverine discharges from the Garonne and Dordogne rivers. By successfully executing complex tidal simulations in fractions of a second during inference, Hydro-PI-GNN achieves a radical reduction in computational time compared to the traditional numerical solver. This research establishes a highly scalable, physically consistent, and open-source methodological framework for deploying high-speed digital twins in vulnerable coastal environments.

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Nomenclature

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Chapter 1

INTRODUCTION

1.1 Background

Coastal and estuarine environments stand among the most dynamic, complex, and hydrologically sensitive regions on Earth. Governed by the intricate interplay of tidal forcing from the ocean, continuous riverine discharge from upstream catchments, and the complex geometry of local bathymetry, free-surface water flows in these transitional zones dictate a broad spectrum of critical environmental and societal outcomes. These include flood inundation dynamics, sediment transport mechanics, salinity intrusion into freshwater aquifers, ecological habitat sustainability, and the safety of maritime navigation [1], [2]. The ability to accurately predict water surface elevation and velocity fields across an estuary is therefore of fundamental and growing importance to coastal engineers, environmental managers, urban planners, and climate scientists alike.

The global threat of climate change has rendered accurate hydrodynamic prediction more urgent than ever before. Projections from the Intergovernmental Panel on Climate Change (IPCC) and numerous subsequent studies indicate that rising mean sea levels, combined with more frequent and intense storm surges and shifting river discharge regimes, will substantially increase the frequency, spatial extent, and severity of coastal flooding worldwide [3], [4], [5]. In macro-tidal estuaries—where natural tidal ranges may exceed several metres and the non-linear coupling between oceanic tides and extreme riverine floods generates highly

complex and unpredictable inundation patterns—even marginal inaccuracies in predicted water levels can have catastrophic consequences for human populations, critical infrastructure, and fragile coastal ecosystems. Consequently, robust flood risk assessment and real-time forecasting systems are imperative.

Historically, physics-based numerical hydrodynamic models have served as the foundational standard for simulating complex estuarine flows. Industry-standard solvers such as Deltares’ Delft3D Flexible Mesh (Delft3D FM) [6], TELEMAC-2D [7], and HEC-RAS 2D [8] discretise the depth-averaged Shallow Water Equations (SWE) on high-resolution structured or unstructured computational meshes. These models simulate tidal propagation, wave dynamics, and current interactions with exceptionally high physical fidelity. When carefully calibrated against empirical field observations, such as tide gauge records and Acoustic Doppler Current Profiler (ADCP) measurements, these models reliably reproduce critical hydrodynamic phenomena, including tidal constituents, tidal asymmetry, and the non-linear interactions between river and ocean forcing. However, their extraordinary computational cost presents a major operational limitation. This cost is primarily driven by the Courant–Friedrichs–Lewy (CFL) condition—a strict numerical stability constraint that restricts the simulation Courant number, forcing the solver to take extremely small time steps on high-resolution meshes. Consequently, running these simulations often requires hours or even days of wall-clock time per tidal cycle on modern multi-core high-performance computing (HPC) clusters. This computational burden constitutes a fundamental bottleneck for applications that demand rapid iteration, such as large ensemble runs for probabilistic forecasting, real-time operational flood warning systems, comprehensive uncertainty quantification, or extensive large-scale climate-impact scenario analysis [9], [10].

In response to these computational challenges, the past decade has witnessed a rapid proliferation of machine learning (ML) and deep learning (DL) methods across the earth and environmental sciences [11], [12], [13]. Various architectures, including traditional Artificial Neural Networks (ANNs), Long Short-Term Memory (LSTM) networks for sequential data, Convolutional Neural Networks (CNNs) for spatial grids, and, most recently, Graph Neural Networks (GNNs) for un-

structured spatial data, have been successfully applied to problems ranging from rainfall-runoff modelling and urban flood mapping to global ocean state forecasting [14], [15]. Once trained offline, these data-driven surrogate models can generate predictions in milliseconds, offering computational speed-up factors of several orders of magnitude relative to classical physics-based solvers. Nevertheless, purely data-driven models suffer from a critical flaw: they are fundamentally agnostic to the governing physical laws of fluid mechanics. Consequently, they are prone to producing non-physical, highly erroneous predictions when deployed in out-of-distribution scenarios or extreme conditions not well-represented in their training data [16], [17].

A principled resolution to this dichotomy between computational speed and physical fidelity is provided by the paradigm of **Physics-Informed Neural Networks (PINNs)**, introduced by Raissi, Perdikaris, and Karniadakis [18]. In the PINN framework, the residuals of the governing partial differential equations (PDEs)—in this case, the Shallow Water Equations—are evaluated via automatic differentiation and embedded directly as penalty terms in the neural network’s training loss function. This innovative approach forces the network to honour both the observed empirical data and the underlying physical conservation laws simultaneously. The result is a predictive model that yields solutions which are physically consistent, highly accurate, and robust even in data-sparse regions or under extrapolated boundary conditions. Extensions of PINNs to solve the one- and two-dimensional Shallow Water Equations have been successfully demonstrated in recent exploratory studies [19], [20], [21], confirming their immense potential for hydrodynamic surrogate modelling.

Despite the success of PINNs, a significant architectural challenge arises when applying them to the highly irregular geometry and complex bathymetry of real-world estuaries. Standard PINN architectures, which typically operate on simple continuous coordinates or regular Cartesian grids, cannot adequately represent the complex, unstructured finite-volume meshes commonly used by state-of-the-art numerical solvers like Delft3D FM. To overcome this, **Graph Neural Networks (GNNs)** [22], [23] are employed. GNNs address this limitation naturally by treat-

ing the computational mesh as a graph structure—where mesh cells function as nodes and shared faces act as edges. This allows the network to perform iterative message passing, exchanging spatial information locally in a topology-aware manner that perfectly mirrors the numerical flux exchanges in finite-volume methods. Recent advances, such as the Data-Guided Finite Volume-Informed Neural Network (FVM-PINN) introduced by Liu [35], explicitly highlight that incorporating finite-volume physics and sparse data guidance is critical to preventing standard strong-form PINNs from failing on complex unstructured grids. Merging the physical rigour and PDE-awareness of PINNs with the geometric flexibility and spatial scalability of GNNs yields the highly advanced **Physics-Informed Graph Neural Network (PI-GNN)** framework [24]. This synthesis forms the core methodological foundation of this thesis, aiming to provide a fast, accurate, and physically consistent surrogate for estuarine hydrodynamics.

1.2 Context

The focal point of this study is the **Gironde estuary**, located in south-western France. Formed by the confluence of the Garonne and Dordogne rivers near the Bec d’Ambès and flowing into the Atlantic Ocean at Pointe de Grave, the Gironde is the largest macro-tidal estuary in Western Europe, covering a surface area of approximately 635 square kilometres. Stretching roughly 75 km in length, the estuary is characterised by extremely dynamic hydrodynamics. It experiences mean spring tidal ranges exceeding 5 m at its mouth, which amplify as they propagate upstream, leading to pronounced non-linear tidal distortion and the generation of significant shallow-water tidal constituents (e.g., M4, M6). Additionally, the estuary receives a highly variable seasonal combined riverine discharge from the Garonne and Dordogne rivers, ranging from less than 200 m³/s during dry summer months to overwhelming peak flows exceeding 5 000 m³/s during winter and spring flood events [1], [2].

The Gironde estuary is a region of immense economic and ecological importance. It supports major shipping and maritime traffic navigating to the inland port of Bordeaux, sustains extensive viticultural and agricultural lowlands along its

banks, and provides critical habitats for numerous marine and avian species. Furthermore, it borders the greater Bordeaux metropolitan area, home to more than 800 000 inhabitants. The combination of dense population, vital infrastructure, and extreme tidal-fluvial interactions makes the region exceptionally vulnerable to complex flooding scenarios, necessitating highly accurate and responsive hydrodynamic monitoring and prediction systems.

To capture the dynamics of this complex system, a high-resolution **Delft3D Flexible Mesh (Delft3D FM)** numerical model of the Gironde estuary was independently developed, calibrated, and validated specifically for this research. This custom-built model encompasses a highly detailed domain of 36 271 unstructured finite-volume cells and 53 224 internal faces, serving as the primary source of high-fidelity ground-truth data for this thesis. The numerical model comprehensively resolves the full estuarine domain—extending from the Atlantic open boundary into the upstream tidal limits of both the Garonne and Dordogne rivers. It has been rigorously calibrated and validated against historical observed water surface elevation and velocity data. The simulation output, precisely recorded at 30-minute intervals over a continuous 265-hour tidal cycle, provides a comprehensive, multidimensional dataset of water surface elevation (η), depth-averaged longitudinal velocity (u), and lateral velocity (v). This dataset forms the foundation against which the proposed PI-GNN surrogate model is trained, constrained, and ultimately tested.

The specific computational challenge that motivates this research is the critical need for a robust **parametric surrogate model**. Unlike simple spatial interpolators, a parametric surrogate must accept not only the fixed spatio-temporal coordinates (x, y, t) and static bathymetric depth (z) as inputs, but also dynamic, time-varying boundary conditions. Specifically, the model must be conditioned on the tidal amplitude at the open ocean boundary (H_{ocean}) and the time-varying river discharges from the Garonne (Q_{Garonne}) and Dordogne (Q_{Dordogne}) rivers. A successfully trained parametric surrogate can then instantly predict the full hydrodynamic state $[\eta, u, v]$ at any spatial node within the estuary, at any given time, for *any* arbitrary combination of these boundary forcings. Crucially, it ac-

compleishes this without requiring a costly re-simulation of the underlying physics-based numerical model. This transformative capability is particularly valuable for enabling real-time operational flood forecasting, conducting ensemble-based probabilistic risk assessments, and exploring diverse, long-term climate-change scenarios with unprecedented computational efficiency.

However, developing such a surrogate for tidal environments presents a profound technical challenge known as **spectral bias**. In the context of deep learning, standard multilayer perceptron (MLP) networks exhibit a strong propensity to learn low-frequency, slowly varying components of a target signal rapidly, while fundamentally struggling to resolve high-frequency, oscillatory components [25]. Because estuarine hydrodynamics are fundamentally dominated by high-frequency tidal oscillations (such as the dominant semi-diurnal ~ 12 h and quarter-diurnal ~ 6 h periods), a standard neural network will fail to capture the complex tidal waves, resulting in overly smoothed, inaccurate predictions. This thesis directly addresses the critical issue of spectral bias through the novel implementation of **Random Fourier Feature (RFF) embeddings**. By projecting the input coordinates into a higher-dimensional periodic space with independently tuned temporal and spatial bandwidths, the RFF encoding effectively allows the PI-GNN to perceive and accurately reconstruct the complex, high-frequency tidal dynamics inherent to the estuarine system.

1.3 Aims and Objectives

The overarching **aim** of this Master’s thesis is to conceptualise, develop, train, and rigorously validate a state-of-the-art parametric Physics-Informed Graph Neural Network (PI-GNN) surrogate model. This model is designed to accurately and instantaneously emulate the complex 2D tidal hydrodynamics of the macro-tidal Gironde estuary. This surrogate learns directly from high-fidelity simulation data generated by the newly developed and calibrated Delft3D FM model, while being explicitly constrained by the governing physical laws of fluid motion. The resulting surrogate must map a comprehensive seven-dimensional input vector $[t, x, y, z, H_{\text{ocean}}, Q_{\text{Garonne}}, Q_{\text{Dordogne}}]$ to predict the full hydrodynamic state vari-

ables $[\eta, u, v]$, perfectly conditioned on dynamic boundary forcings, thereby bypassing the need for computationally prohibitive numerical simulations.

To achieve this overarching aim, the study pursues the following specific **objectives**:

1. **Numerical Modelling and Data Curation:** To independently develop and calibrate a high-fidelity Delft3D FM numerical model of the Gironde estuary. Subsequently, to meticulously extract, process, and normalize a massive training dataset from its simulations. This includes processing the complex unstructured mesh geometry, extracting cell-centred bathymetric depths, and preparing the time-series data of water surface elevation and depth-averaged velocity fields across the entire 265-hour simulation period, ensuring optimal formatting for graph-based deep learning.
2. **Architecture Design with Spectral Bias Mitigation:** To design and engineer a robust PI-GNN architecture that explicitly overcomes spectral bias. This involves integrating Random Fourier Feature (RFF) embeddings—utilising highly specific and decoupled temporal ($\sigma_t = 30$) and spatial ($\sigma_s = 1$) bandwidths—with a deep, six-layer message-passing Multi-Layer Perceptron (comprising 512 hidden units per layer with SiLU activations). This network will operate directly on the unstructured finite-volume cell graph derived from the Gironde computational mesh.
3. **Physics-Informed Loss Formulation:** To mathematically formulate and programmatically implement a comprehensive, autograd-based physics-informed loss function. This function must explicitly encode the complete 2D Shallow Water Equations, encompassing mass conservation, x-momentum, and y-momentum. Crucially, it must incorporate complex bed-slope corrections based on the estuary’s bathymetry. This physics loss will be optimized alongside traditional data fidelity (Mean Squared Error), boundary condition, and initial condition terms, managed via an adaptive, multi-stage curriculum training schedule to ensure stable convergence.
4. **Model Validation and Computational Acceleration:** To comprehen-

sively validate the predictive accuracy and physical consistency of the fully trained PI-GNN surrogate on a temporally held-out test dataset (representing the final 20% of the simulation, or approximately 55 hours of entirely unseen tidal forcing). Furthermore, to explicitly quantify the dramatic reduction in simulation inference time achieved by the surrogate relative to the traditional numerical solver. Performance will be quantified using standard hydrological metrics, including Root Mean Square Error (RMSE), Nash–Sutcliffe Efficiency (NSE), and the coefficient of determination (R^2), evaluated rigorously at eight designated observation stations and five strategically located interior nodes across the estuary.

5. **Assessment of Parametric Generalisation:** To critically assess the parametric generalisation capabilities of the surrogate model by evaluating its predictive accuracy and stability when subjected to novel combinations of tidal amplitudes and riverine discharges that were not encountered during the model’s training phase. This objective aims to definitively demonstrate the model’s operational utility as a high-speed digital twin for practical scenario analysis and flood forecasting.

1.4 Significance, Scope and Definitions

Significance

This research constitutes a significant advancement at the intersection of computational coastal hydrology and applied scientific machine learning. It makes several original and impactful contributions to the field:

- **Large-Scale Parametric Emulation:** It presents one of the first successful implementations of a fully **parametric PI-GNN surrogate model** for modelling 2D tidal hydrodynamics on a real, large-scale, highly dynamic macro-tidal estuary. The successful handling of an unstructured finite-volume mesh comprising more than 36 000 cells demonstrates the scalability of graph-based methods for real-world coastal engineering applications.
- **Advanced Physics Integration:** It demonstrates the successful practical

implementation of highly complex, **autograd-based 2D SWE physics residuals** within a graph-structured deep learning framework. By successfully encoding mass conservation and momentum equations, including critical bathymetric bed-slope corrections computed via Green–Gauss gradient reconstruction, it pushes the boundaries of how much physical realism can be embedded in neural networks for irregular domains.

- **Overcoming Spectral Bias in Tidal Modelling:** It introduces and validates a novel **dual-bandwidth Random Fourier Feature encoding** scheme that is specifically tailored to the distinct temporal (rapid tidal) and spatial (large-scale estuarine) length scales of coastal problems. This directly and effectively addresses the spectral bias phenomenon that typically causes standard PINNs to fail when modelling highly oscillatory tidal signals.
- **Transformative Computational Acceleration:** It definitively proves that complex, physics-constrained tidal hydrodynamics can be simulated in fractions of a second during inference. By drastically reducing the computational time compared to traditional CPU-bound numerical solvers, it paves the way for real-time flood warning systems and large-scale ensemble forecasting.
- **Open-Source Methodology:** It provides a fully **reproducible and open-source training pipeline**, which has been thoroughly tested and optimized on GPU-accelerated computing platforms (such as Kaggle and local CUDA environments). This pipeline is highly adaptable and can be seamlessly applied to other complex estuarine or coastal basins worldwide, provided they have baseline calibrated numerical simulation datasets.

Scope

To ensure focus and computational feasibility, the scope of this study is specifically limited to the following parameters:

- **Physical Equations:** The modelling is restricted strictly to **depth-averaged, 2D free-surface hydrodynamics** as governed by the Shallow Water Equa-

tions. Complex three-dimensional flow structures, advanced turbulence closure models (e.g., $k - \epsilon$), and density stratification effects (baroclinic flows) are explicitly excluded from this analysis.

- **Geographical Domain:** The study is exclusively focused on the **Gironde estuary** in France. The simulations utilise data generated by the Delft3D FM numerical model developed as part of this thesis, applying a constant, uniform Manning’s bed roughness coefficient of $n = 0.019 \text{ s m}^{-1/3}$ across the entire domain.
- **Target Variables:** The surrogate model is trained to predict **hydrodynamic state variables only**: specifically, water surface elevation (η), depth-averaged eastward velocity (u), and depth-averaged northward velocity (v). Phenomena such as sediment transport, morphological evolution, wave action, and salinity or pollutant intrusion are beyond the current scope.
- **Temporal Domain:** The analysis is based on a continuous **265-hour simulation period**, which captures multiple distinct tidal cycles. This dataset is rigorously split using an 80/20 chronological ratio, resulting in a training set of approximately 210 hours and a temporally distinct, held-out test set of approximately 55 hours for uncompromised validation.

Definitions

To ensure clarity and precision, key technical terms utilized extensively throughout this thesis are defined below:

Shallow Water Equations (SWE): The depth-averaged, vertically integrated form of the Navier–Stokes equations. They govern free-surface fluid flows in scenarios where the horizontal length scale of the fluid motion greatly exceeds the vertical water depth, describing the fundamental conservation of mass and depth-averaged momentum in two horizontal spatial dimensions.

Physics-Informed Neural Network (PINN): An advanced class of neural networks trained by minimizing a composite loss function. This function uniquely incorporates traditional data fidelity terms (to match observations) alongside

PDE residual terms. These residual terms act as soft penalties for any violations of the underlying governing physical equations, computed efficiently using automatic differentiation within the neural network framework.

Graph Neural Network (GNN): A highly specialized category of deep learning architectures designed to operate directly on arbitrarily structured graph data. GNNs function through iterative message passing and aggregation mechanisms between connected nodes. This allows the network to learn complex representations that explicitly respect the geometric topology and connectivity of the underlying graph—in this context, the unstructured computational mesh of the estuary.

PI-GNN (Physics-Informed Graph Neural Network): A novel hybrid deep learning architecture that synergistically combines the rigorous physical constraint enforcement characteristic of PINNs with the unparalleled geometric flexibility and spatial awareness of GNNs. It is specifically designed for highly accurate, physics-consistent surrogate modelling on complex, irregular computational meshes.

Surrogate Model: A highly optimized, computationally inexpensive mathematical or algorithmic approximation of a complex, high-fidelity numerical simulator. By learning complex input-output mappings from pre-computed simulation data, a surrogate is capable of producing highly accurate predictions in a tiny fraction of the time required by the original, computationally heavy physics-based solver.

Random Fourier Feature (RFF) Embedding: A sophisticated mathematical technique used to map low-dimensional input coordinates (such as time t or spatial coordinates x, y) into a much higher-dimensional feature space using specifically tuned, random sinusoidal projections. This embedding is crucial for enabling deep neural networks to accurately learn high-frequency functions, effectively overcoming the inherent limitation known as spectral bias [25].

Parametric Surrogate: An advanced type of surrogate model that is designed

to condition its predictions dynamically based on explicit, variable boundary conditions or forcing parameters (in this study, ocean tidal amplitude and various upstream riverine discharges). This parametric capability enables rapid, on-the-fly prediction across a virtually infinite family of different environmental scenarios without the need for model retraining.

Curriculum Learning: A strategic, phased machine learning training paradigm.

In curriculum learning, the difficulty, complexity, or composition of the learning task is deliberately and progressively increased during the training process. In the context of PINNs, this often involves initially training the network to fit only the empirical data without physics constraints, and then gradually and systematically introducing the complex PDE penalties as the model’s accuracy improves [26].

Manning’s Roughness Coefficient (n): An essential empirical parameter utilised

in the SWE friction term (specifically, the Manning–Strickler formulation). It represents the physical resistance to fluid flow offered by the roughness of the channel bed. For this study, it is defined as a uniform constant of $n = 0.019 \text{ s m}^{-1/3}$ across the entire estuarine domain.

Nash–Sutcliffe Efficiency (NSE): A robust, dimensionless statistical metric

(ranging from $-\infty$ to 1) that is universally employed in hydrology to rigorously quantify the predictive skill of a model. An NSE value of 1 denotes a perfect match between model predictions and observed data, an NSE of 0 indicates that the model predictions are only as accurate as the mean of the observed data, and an $\text{NSE} < 0$ indicates that the model is fundamentally flawed and performs worse than simply using the observed mean.

1.5 Thesis Outline

The remainder of this thesis is structured to provide a comprehensive and logical progression of the research, organized as follows:

Chapter 2 (Literature Review) provides an exhaustive and critical review of the existing body of knowledge. It systematically explores: (i) the foundational

principles of traditional numerical hydrodynamic modelling and its historical application to complex estuarine systems; (ii) the rapid theoretical development and practical application of Physics-Informed Neural Networks (PINNs) for solving complex PDEs in fluid mechanics; (iii) the emergence of Graph Neural Networks (GNNs) for analyzing unstructured-mesh hydrodynamics and their critical role in accelerating traditional numerical solvers; and (iv) state-of-the-art surrogate modelling strategies for real-time coastal flood mapping and tidal forecasting. The chapter concludes by synthesizing a clear summary of the identified critical research gaps that directly motivate the objectives of the present study.

Chapter 3 (Materials and Methodology) provides a deeply detailed exposition of the research materials and the proposed methodologies. It begins by describing the complex geophysical and hydrodynamic characteristics of the study area (the Gironde estuary). It then details the independent development and configuration of the Delft3D FM numerical model, the rigorous calibration procedure applied against field observations, and the precise nature of the simulation dataset generated for neural network training and validation. Following this, the chapter meticulously details the architecture of the proposed PI-GNN surrogate. This includes a deep dive into the Random Fourier Feature encoding strategy, the design of the message-passing graph neural network structure, the complex mathematical formulation of the autograd-based physics-informed loss function, the rationale behind the multi-stage curriculum training strategy, and the specifics of the GPU-accelerated computational implementation.

Chapter 4 (Results) presents the comprehensive findings of the model training and validation phases. It includes detailed analyses of the training convergence curves and explicitly reports quantitative performance metrics (including RMSE, NSE, and R^2) evaluated at specific observation stations and across interior nodes for both the training phase and the critical held-out test period. Furthermore, it presents vivid spatial comparisons of water-level and velocity fields, alongside descriptions of the animated tidal simulation outputs generated by the fully trained PI-GNN surrogate.

Chapter 5 (Analysis and Discussion) provides a critical and in-depth analysis

of the results. It interprets the PI-GNN surrogate’s performance explicitly in relation to the research gaps identified in Chapter 2. It rigorously examines the individual contributions and impacts of the various components of the physics loss terms, compares the advanced PI-GNN model against baseline purely data-driven models to highlight the value of physics integration, and extensively discusses the broader real-world implications of the research for operational tidal forecasting systems and long-term climate-change impact assessments.

Chapter 6 (Conclusions) serves as the final synthesis of the thesis. It concisely summarises the principal findings of the entire study, clearly restates the original contributions made to the scientific community, candidly acknowledges the inherent limitations and boundaries of the present work, and provides thoughtful recommendations and clear directions for future research endeavours. These recommendations include exploring multi-fidelity training paradigms, investigating domain adaptation techniques to apply the model to other estuaries globally, and extending the current architecture to encompass coupled hydrodynamic and sediment transport modelling.

Chapter 2

LITERATURE REVIEW

2.1 General

The literature review chapter demonstrates a thorough knowledge of the interdisciplinary areas fundamental to this thesis, specifically coastal hydrodynamic modelling, deep learning, and physics-informed surrogate frameworks. The primary objective is to critically evaluate the progression from traditional physics-based numerical solvers to contemporary machine learning approaches in fluid mechanics, highlighting the strengths and inherent limitations of each paradigm. By synthesizing recent advancements in Physics-Informed Neural Networks (PINNs), Graph Neural Networks (GNNs), and spectral bias mitigation techniques, this chapter provides the foundational arguments to support the study's focus: the development of a fully parametric Physics-Informed Graph Neural Network (PI-GNN) for estuarine tidal hydrodynamics.

2.2 Historical Background

2.2.1 Traditional Numerical Hydrodynamic Modelling

Historically, the prediction of estuarine and coastal hydrodynamics has relied exclusively on physics-based numerical models. These models fundamentally solve the Navier–Stokes equations, most commonly simplified to the depth-averaged

Shallow Water Equations (SWE) for free-surface flows where horizontal scales vastly exceed the vertical scale [27]. Industry-standard solvers such as Delft3D Flexible Mesh (Delft3D FM) [6] and TELEMAC-2D [7] employ advanced numerical schemes, such as the Finite Volume Method (FVM), to discretise the spatial domain into highly irregular, unstructured meshes. These unstructured grids allow for high-resolution representation of complex coastlines, deep navigational channels, and highly variable bathymetry, which are critical for capturing non-linear tidal propagation and riverine interactions.

When properly calibrated against field measurements, these numerical models exhibit remarkable accuracy and physical fidelity [8], [9]. However, their profound limitation lies in their computational expense. The necessity to satisfy strict stability criteria (e.g., the Courant–Friedrichs–Lewy condition) forces these models to operate on very small time steps, often requiring hours or days of high-performance computing resources to simulate a single tidal cycle. As the global imperative to assess climate-driven hydrological extremes grows—driven by rising sea levels and intensified storm surges [3], [4], [5]—the prohibitive cost of these traditional models renders them fundamentally unsuited for large-scale probabilistic risk assessments, real-time flood warning systems, and rapid scenario exploration.

2.2.2 The Rise of Data-Driven Machine Learning in Hydrology

To circumvent the computational bottlenecks of traditional solvers, the past decade has seen a paradigm shift towards data-driven surrogate modelling, spurred by broader advancements in deep learning [11], [12]. In hydrology and water resources, deep neural networks—including Long Short-Term Memory (LSTM) networks and Convolutional Neural Networks (CNNs)—have been extensively deployed for tasks ranging from rainfall-runoff prediction to flood inundation mapping [13], [14], [15]. For instance, Mo et al. [28] successfully applied deep learning for predicting estuarine and coastal storm surges, significantly accelerating prediction times compared to numerical simulators.

These pure machine learning models learn complex input-output mappings di-

rectly from historical data or numerical simulation archives. Once trained, they can execute predictions in milliseconds, offering a transformative acceleration in computational speed [10], [29]. However, these models operate as “black boxes” that are fundamentally agnostic to the physical laws governing fluid dynamics. Consequently, they often struggle with spatial generalization, fail to ensure physical conservation (such as mass and momentum balance), and are highly susceptible to producing catastrophic, non-physical predictions when presented with out-of-distribution boundary conditions [16], [17].

2.2.3 Physics-Informed Neural Networks and Gradient Pathologies

A revolutionary resolution to the dichotomy between physical rigor and computational speed was the introduction of Physics-Informed Neural Networks (PINNs) by Raissi et al. [18]. PINNs seamlessly integrate observational data with the governing partial differential equations (PDEs) of a system. By leveraging automatic differentiation, the residuals of the PDEs (e.g., the SWE) are evaluated during the training process and added as penalty terms to the loss function. This constrains the neural network to explore only the hypothesis space that adheres to known physical laws.

Recent literature demonstrates the immense potential of PINNs for hydrodynamic applications. Sun et al. [30] demonstrated surrogate modeling for fluid flows using physics-constrained learning without requiring massive simulation datasets. More specifically, studies have begun extending the PINN framework to the Shallow Water Equations. Yin et al. [20] successfully solved the 1D unsteady SWE using a PINN approach, while Tian et al. [19] and Dazzi et al. [21] expanded this to 2D SWE domains, explicitly incorporating complex topographical and bathymetric source terms.

Despite these successes, PINNs frequently suffer from severe training instabilities when applied to complex multi-physics problems. Wang et al. [31] highlighted that PINNs are highly susceptible to “gradient pathologies,” where the gradient of the loss function with respect to the physical residuals completely dominates

the gradients of the data or boundary conditions, leading to failure in convergence. This underscores the necessity for advanced optimization strategies and careful loss weighting when developing robust surrogate models.

2.2.4 Graph Neural Networks and Unstructured Meshes

A critical architectural limitation of standard PINNs is their reliance on Multi-Layer Perceptrons (MLPs) mapping continuous spatial coordinates (x, y) or Convolutional Neural Networks (CNNs) operating on regular Cartesian grids. This renders them fundamentally incompatible with the unstructured, highly heterogeneous finite-volume meshes that characterize state-of-the-art hydrodynamic models like Delft3D FM.

To bridge this structural gap, researchers have increasingly adopted Graph Neural Networks (GNNs) [22], [23]. GNNs treat the computational mesh naturally as a graph, where finite-volume cells act as nodes and the shared faces between them act as edges. By employing iterative message passing, GNNs learn complex spatial representations that explicitly respect the unstructured topology of the domain. In a seminal work, Pfaff et al. [32] demonstrated that graph networks can successfully learn to simulate highly complex, mesh-based physics across fluid and solid mechanics. Building on this, recent applications by Maier et al. [33] and Lino et al. [34] have confirmed that GNNs can effectively emulate numerical fluxes across irregular coastal grids. Crucially, Liu [35] introduced the Data-Guided Finite Volume-Informed Neural Network (FVM-PINN), emphasizing that strong-form PINNs fail on unstructured shallow water grids unless formulated with finite-volume physics and guided by sparse observational data. The fusion of GNNs with physics-informed training—yielding the Physics-Informed Graph Neural Network (PI-GNN)—has recently been proposed as the optimal architecture for flood forecasting on complex terrains, perfectly marrying geometric flexibility with physical adherence [24], [36].

2.2.5 Mitigating Spectral Bias in Tidal Dynamics

A foundational challenge in applying deep learning to estuarine hydrodynamics is the phenomenon of “spectral bias.” As established by Rahaman et al. [37], neural networks inherently prioritize learning low-frequency, smoothly varying components of a signal, whilst struggling profoundly to resolve high-frequency, highly oscillatory dynamics.

In macrotidal estuaries, hydrodynamics are utterly dominated by high-frequency semi-diurnal and quarter-diurnal tidal oscillations. Standard neural networks tend to produce overly smoothed, inaccurate representations of these tidal waves. To counteract this, Tancik et al. [25] demonstrated that mapping low-dimensional inputs (such as time and space) into a high-dimensional periodic space using Random Fourier Feature (RFF) embeddings empowers networks to overcome spectral bias, enabling the accurate reconstruction of high-frequency functions. Applying RFF positional encoding to hydrodynamic PINNs is an emerging and highly critical requirement for accurately modelling tidal signals.

2.2.6 The Gironde Estuary as a Complex Test Bed

The Gironde estuary in south-western France presents a highly challenging and scientifically relevant environment for testing advanced hydrodynamic models. As the largest macrotidal estuary in Western Europe, it is characterized by massive tidal ranges, severe non-linear tidal asymmetry, and highly variable riverine discharges from the Garonne and Dordogne rivers [1], [2].

Previous studies on the Gironde have extensively documented the complex interactions between tidal forcing and riverine flow, highlighting its intricate sediment transport mechanisms and cohesive sediment suspension dynamics [38]. Because of its extreme hydrodynamic variability, traditional numerical methods are typically required to model it accurately [39]. Its geometric complexity, vast physical scale, and extreme non-linear tidal dynamics make the Gironde an ideal, rigorous proving ground for the proposed PI-GNN surrogate model.

2.3 Summary and Implications

The comprehensive review of the literature reveals a clear trajectory in hydrodynamic modelling: progressing from computationally intensive physics-based solvers, through rapid but physically inconsistent data-driven models, and now converging upon physically constrained machine learning frameworks. To synthesize this progression, Table 2.1 provides a direct comparison between the current state-of-the-art surrogate modelling approaches and the methodology proposed in this thesis.

Table 2.1: Summary of state-of-the-art surrogate modelling methodologies compared to the proposed Hydro-PI-GNN framework.

Study / Reference	Methodology	Application Domain	Mesh / Grid	Physical Consistency	Spectral Bias Addressed?
Sit et al. (2020) [14]	Deep Learning (GCN)	Streamflow / Hydrology	Graph/Network	No (Data-Driven)	No
Mo et al. (2020) [28]	Deep Learning (CNN/LSTM)	Coastal Storm Surges	Regular Grid	No (Data-Driven)	No
Raissi et al. (2019) [18]	PINN (Original)	General Fluid PDEs	Continuous 1D/2D	Yes (PDEs)	No
Tancik et al. (2020) [25]	Fourier Features (MLP)	1D/2D Signals	Continuous	No (Data-Driven)	Yes (RFF)
Yin et al. (2024) [20]	Physics-Informed Neural Network (PINN)	1D Shallow Water	Continuous 1D	Yes (SWE)	No
Dazzi et al. (2024) [21]	Advanced PINN	2D Shallow Water	Regular Cartesian	Yes (SWE + Topo)	No
Tian et al. (2025) [19]	Advanced PINN	2D Shallow Water	Regular Cartesian	Yes (SWE + Topo)	No
Pfaff et al. (2021) [32]	Graph Neural Networks (GNN)	General Fluid Mechanics	Unstructured	No (Implicit)	No
Maier et al. (2023) [33]	Graph Neural Networks (GNN)	Coastal Flows	Irregular / Mesh	No (Implicit)	No
Liu (2026) [35]	FVM-PINN	2D Shallow Water	Finite Volume Mesh	Yes (FVM-SWE)	No
Taghizadeh et al. (2025) [24]	PI-GNN	Flood Forecasting	Graph-based	Yes	No
This Study (Hydro-PI-GNN)	PI-GNN with RFF Encoding	Macrotidal Estuary (2D SWE)	Unstructured (Delft3D FM)	Yes (Autograd SWE)	Yes (Dual-Band RFF)

Several critical research gaps have been identified in the current state-of-the-art:

1. Purely data-driven models (e.g., [28]) cannot guarantee physical consistency, limiting their operational reliability during extreme, unseen weather events.
2. While standard PINNs (e.g., [19]) have demonstrated success in solving the

Shallow Water Equations, they are largely restricted to continuous coordinate mapping or regular grids, rendering them incompatible with the unstructured finite-volume meshes utilized by modern estuarine models like Delft3D FM.

3. Standard deep learning architectures suffer from well-documented spectral bias [37], routinely failing to capture the high-frequency tidal oscillations that define macro-tidal estuarine dynamics without specific feature embedding [25].
4. There is a distinct lack of fully parametric PI-GNN surrogates capable of modelling large-scale, real-world estuaries while being explicitly conditioned on highly variable, dynamic boundary conditions (such as varying ocean tides and upstream river discharges).

These identified gaps directly form the implications and foundational justification for this thesis. To advance the field of real-time coastal forecasting, it is necessary to move beyond simple grid-based PINNs. The literature strongly supports the development of a hybrid approach: merging the topological awareness of Graph Neural Networks [32] with the physical rigor of autograd-based Shallow Water Equation residuals. Furthermore, the integration of Random Fourier Feature (RFF) embeddings is implied as a critical necessity to overcome the spectral bias inherent in tidal modelling. By independently developing and calibrating a Delft3D FM numerical model of the highly dynamic Gironde estuary, and subsequently training a parametric PI-GNN architecture upon its unstructured mesh, this study directly addresses the limitations of current methodologies. This approach promises to deliver a digital twin capable of unprecedented computational speed without sacrificing the physical fidelity demanded by modern coastal engineering and climate impact assessment.

Chapter 3

MATERIALS AND METHODOLOGY

3.1 Materials

3.1.1 Basic Research Materials

3.2 Experimental investigation/methodology

3.2.1 Methodology

3.2.2 Research Design

3.3 Procedure and Timeline

3.4 Analysis

3.5 Ethics and Limitations

Chapter 4

RESULTS

4.1 Topic 1

4.1.1 Case 1

4.1.2 Case 2

4.2 Topic 2

Chapter 5

ANALYSIS/DISCUSSION

5.1 General

[This chapter contains a full discussion, interpretation, and evaluation of the results with reference to the literature.]

Chapter 6

CONCLUSIONS

6.1 General

6.2 Key Findings

6.3 Limitation of the Study

6.4 Practical Implication

6.5 Recommendation for Further Study

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Appendix A

[Title of First Appendix]

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